RT-DocLayout: Real-Time End-to-End Document Layout Analysis with Reading Order in the Wild — Supplementary Material —

Cheng Cui *, Tingquan Gao *, Xueqing Wang, Changda Zhou, Hongen Liu, Ting Sun, Yubo Zhang, Zelun Zhang, Jiaxuan Liu, Manhui Lin, Yue Zhang, Suyin Liang, Yiqing Xiang, and Yi Liu **

PaddlePaddle Team, Baidu Inc., Beijing, China paddleocr@baidu.com

1 Inference Efficiency

To evaluate the practical deployment viability of our proposed RT-DocLayout, we conduct a comprehensive benchmark of inference efficiency against several state-of-the-art (SOTA) document layout analysis models, including DocLayout-YOLO [7], MinerU2.5 [5], Dolphin-v2 [3], and PP-DocLayoutV2 [2].

Table 1: Average FPS of various layout analysis models.

Model	Backend	Average FPS at Different Batch Sizes ↑					
		1	2	4	8	16	32
DocLayout-YOLO [7]	PyTorch 2.10.0	40.9	89.0	101.0	102.4	102.7	107.4
MinerU2.5 [5]	vLLM 0.10.2	0.4	0.6	0.9	1.2	1.9	2.4
Dolphin-v2 [3]	vLLM 0.10.2	0.3	0.4	0.5	0.5	0.7	0.9
PP-DocLayoutV2 [2]	PyTorch 2.10.0	24.9	44.3	80.6	87.7	98.2	110.9
RT-DocLayout (Ours)	PyTorch 2.10.0	**27.6**	**51.8**	**82.7**	**105.4**	**129.1**	**132.1**

All efficiency evaluations are conducted on a workstation equipped with an NVIDIA A100 (80GB) GPU and an Intel(R) Xeon(R) Platinum 8350C CPU @ 2.60GHz. We adopt the OmniDocBench v1.5 [6] dataset, comprising 1,355 diverse document images, as the test benchmark. For a consistent and fair comparison, each model is evaluated using its native inference backend (PyTorch or vLLM) across varying batch sizes ($BS \in \{1, 2, 4, 8, 16, 32\}$) to simulate a range of real-world throughput demands.

As shown in Table 1, RT-DocLayout achieves superior throughput across all batch sizes. At $BS = 32$, RT-DocLayout reaches 132.1 FPS, outperforming DocLayout-YOLO [7] (107.4 FPS) and PP-DocLayoutV2 [2] (110.9 FPS) by

* Equal contribution.
** Corresponding author.

23.0% and 19.1%, respectively. This gain stems from our unified end-to-end architecture, which integrates element classification, localization, and reading order prediction into a single forward pass, eliminating redundant feature extraction and costly post-processing heuristics. Compared to VLM-based approaches like MinerU2.5 [5] and Dolphin-v2 [3], which are constrained by heavy autoregressive decoding, RT-DocLayout achieves a 55× to 147× speedup at $BS = 32$. While VLMs struggle to exceed 3 FPS even with vLLM acceleration, our model offers a significantly more scalable and cost-effective solution for industrial-grade, large-scale document parsing.

In conclusion, RT-DocLayout represents a significant advancement by achieving comprehensive leadership in both parsing accuracy and inference throughput, making it the most balanced solution for high-performance production environments.

2 Robustness Analysis across Diverse Document Distortions

To validate the effectiveness and generalizability of our proposed RT-DocLayout, we conduct extensive cross-module benchmarks on Real5-OmniDocBench [8]. Specifically, we evaluate DocLayout-YOLO [7], MinerU2.5 [5], Dolphin-v2 [3], PP-DocLayoutV2 [2], and our RT-DocLayout, each integrated with various downstream recognition systems. Tables 2 to 6 summarize the comprehensive performance across different document degradation scenarios.

In challenging scenarios involving Warping and Skew, traditional and baseline models exhibit significant performance degradation. For instance, in the Warping test (Table 3), Dolphin-v2 [3] suffers a sharp decline in Overall scores (dropping to the 40–60 range). In contrast, RT-DocLayout demonstrates remarkable stability, maintaining an Overall score of 91.09. Similarly, in the Skew scenario (Table 6), where performance of DocLayout-YOLO [7] is constrained (yielding an Overall score of ∼50.06), RT-DocLayout achieves a superior 91.72. This resilience is further evidenced by the Reading Order Edit metric: our method significantly outperforms DocLayout-YOLO [7] paired with XYCut post-processing (e.g., 0.062 vs. 0.583 in the Skew subset). This validates that our model effectively models complex structural relationships even under non-linear geometric deformations.

Furthermore, RT-DocLayout exhibits exceptional robustness to environmental noise in Scanning, Screen-Photography, and Illumination tasks. It consistently provides high-quality region crops for specialized recognition heads, as reflected by the leading Formula CDM and Table TEDS scores. Across all five benchmarks, the integration of RT-DocLayout and PaddleOCR-VL-1.5-0.9B consistently yields the state-of-the-art (SOTA) performance, establishing our module as a versatile, "plug-and-play" solution for robust document parsing in real-world applications.

Table 2: Comparison of different layout analysis modules combined with various recognition systems on Real5-OmniDocBench Scanning subset. As DocLayout-YOLO does not output reading order, we employ the XYCut algorithm as a post-processing.

Layout Analysis	Recognition Method	Overall↑	Text^{Edit}↓	Formula^{CDM}↑	Table^{TEDS}↑	Reading Order^{Edit}↓
DocLayout-YOLO [7]	MonkeyOCR [4]	84.09	0.137	84.09	81.89	0.179
	MinerU2.5 [5]	85.00	0.142	85.13	84.11	0.182
	Dolphin-v2 [3]	83.40	0.156	82.56	83.19	0.186
	PaddleOCR-VL-1.5-0.9B [1]	85.57	0.133	85.69	84.36	0.180
MinerU2.5 [5]	MonkeyOCR [4]	88.70	0.054	86.35	85.10	0.056
	MinerU2.5 [5]	88.37	0.065	85.59	86.06	0.066
	Dolphin-v2 [3]	87.20	0.076	82.23	86.95	0.066
	PaddleOCR-VL-1.5-0.9B [1]	89.74	0.050	85.88	88.30	0.054
Dolphin-v2 [3]	MonkeyOCR [4]	87.86	0.062	86.18	83.58	0.063
	MinerU2.5 [5]	87.70	0.071	85.16	85.05	0.074
	Dolphin-v2 [3]	85.72	0.085	81.84	83.85	0.076
	PaddleOCR-VL-1.5-0.9B [1]	88.66	0.059	86.38	85.52	0.062
PP-DocLayoutV2 [2]	MonkeyOCR [4]	90.47	0.046	89.07	86.87	0.052
	MinerU2.5 [5]	90.98	0.049	89.32	88.54	0.055
	Dolphin-v2 [3]	89.46	0.068	86.82	88.35	0.062
	PaddleOCR-VL-1.5-0.9B [1]	92.04	0.039	89.81	90.22	0.048
RT-DocLayout	MonkeyOCR [4]	91.73	0.042	91.04	88.38	0.047
	MinerU2.5 [5]	91.74	0.045	90.35	89.35	0.049
	Dolphin-v2 [3]	90.36	0.064	88.06	89.39	0.059
	PaddleOCR-VL-1.5-0.9B [1]	**93.25**	**0.037**	**92.33**	**91.11**	**0.046**

Table 3: Comparison of different layout analysis modules combined with various recognition systems on Real5-OmniDocBench Warping subset. As DocLayout-YOLO does not output reading order, we employ the XYCut algorithm as a post-processing.

Layout Analysis	Recognition Method	Overall↑	Text^{Edit}↓	Formula^{CDM}↑	Table^{TEDS}↑	Reading Order^{Edit}↓
DocLayout-YOLO [7]	MonkeyOCR [4]	74.81	0.255	81.30	68.59	0.342
	MinerU2.5 [5]	72.92	0.301	79.70	69.20	0.364
	Dolphin-v2 [3]	73.59	0.271	77.73	70.17	0.359
	PaddleOCR-VL-1.5-0.9B [1]	74.70	0.269	80.47	70.49	0.349
MinerU2.5 [5]	MonkeyOCR [4]	84.10	0.102	83.65	78.88	0.089
	MinerU2.5 [5]	82.85	0.153	84.45	79.44	0.116
	Dolphin-v2 [3]	82.72	0.122	79.36	81.02	0.097
	PaddleOCR-VL-1.5-0.9B [1]	83.92	0.119	81.90	81.80	0.093
Dolphin-v2 [3]	MonkeyOCR [4]	58.41	0.305	75.45	30.23	0.184
	MinerU2.5 [5]	45.38	0.461	51.29	30.96	0.357
	Dolphin-v2 [3]	58.08	0.359	54.06	56.09	0.246
	PaddleOCR-VL-1.5-0.9B [1]	52.58	0.331	61.16	29.71	0.196
PP-DocLayoutV2 [2]	MonkeyOCR [4]	85.90	0.083	87.70	78.34	0.091
	MinerU2.5 [5]	83.88	0.134	86.17	78.83	0.116
	Dolphin-v2 [3]	84.54	0.105	85.41	78.67	0.101
	PaddleOCR-VL-1.5-0.9B [1]	86.00	0.095	86.89	80.56	0.097
RT-DocLayout	MonkeyOCR [4]	89.38	0.058	89.77	84.20	0.069
	MinerU2.5 [5]	88.96	0.079	90.68	84.05	0.075
	Dolphin-v2 [3]	88.47	0.076	87.10	85.90	0.078
	PaddleOCR-VL-1.5-0.9B [1]	**91.09**	**0.053**	**90.59**	**87.95**	**0.064**

Table 4: Comparison of different layout analysis modules combined with various recognition systems on Real5-OmniDocBench Illumination subset. As DocLayout-YOLO does not output reading order, we employ the XYCut algorithm as a post-processing.

Layout Analysis	Recognition Method	Overall↑	Text^{Edit}↓	Formula^{CDM}↑	Table^{TEDS}↑	Reading Order^{Edit}↓
DocLayout-YOLO [7]	MonkeyOCR [4]	81.99	0.167	82.92	79.73	0.230
	MinerU2.5 [5]	82.68	0.177	84.74	80.95	0.237
	Dolphin-v2 [3]	81.00	0.184	80.52	80.84	0.241
	PaddleOCR-VL-1.5-0.9B [1]	83.65	0.163	84.24	83.00	0.228
MinerU2.5 [5]	MonkeyOCR [4]	87.63	0.064	84.10	85.17	0.065
	MinerU2.5 [5]	88.02	0.076	86.34	85.33	0.072
	Dolphin-v2 [3]	86.17	0.086	81.39	85.75	0.077
	PaddleOCR-VL-1.5-0.9B [1]	89.29	0.059	86.06	87.74	0.059
Dolphin-v2 [3]	MonkeyOCR [4]	68.31	0.200	77.62	47.28	0.128
	MinerU2.5 [5]	58.54	0.312	58.45	48.36	0.252
	Dolphin-v2 [3]	64.03	0.269	52.72	66.26	0.192
	PaddleOCR-VL-1.5-0.9B [1]	66.15	0.201	69.40	49.09	0.123
PP-DocLayoutV2 [2]	MonkeyOCR [4]	88.86	0.053	86.77	85.08	0.060
	MinerU2.5 [5]	89.06	0.064	88.23	85.35	0.069
	Dolphin-v2 [3]	87.97	0.075	85.73	85.65	0.073
	PaddleOCR-VL-1.5-0.9B [1]	90.80	0.048	89.83	87.33	0.056
RT-DocLayout	MonkeyOCR [4]	89.72	0.055	88.76	85.86	0.058
	MinerU2.5 [5]	90.31	0.062	90.80	86.31	0.066
	Dolphin-v2 [3]	89.39	0.073	87.50	88.00	0.070
	PaddleOCR-VL-1.5-0.9B [1]	**92.11**	**0.046**	**91.72**	**89.22**	**0.052**

Table 5: Comparison of different layout analysis modules combined with various recognition systems on Real5-OmniDocBench Screen-Photography subset. As DocLayout-YOLO does not output reading order, we employ the XYCut algorithm as a post-processing.

Layout Analysis	Recognition Method	Overall↑	Text^{Edit}↓	Formula^{CDM}↑	Table^{TEDS}↑	Reading Order^{Edit}↓
DocLayout-YOLO [7]	MonkeyOCR [4]	76.89	0.219	79.39	73.17	0.264
	MinerU2.5 [5]	77.38	0.232	80.76	74.57	0.272
	Dolphin-v2 [3]	75.90	0.229	77.15	73.49	0.268
	PaddleOCR-VL-1.5-0.9B [1]	77.67	0.209	79.96	73.98	0.261
MinerU2.5 [5]	MonkeyOCR [4]	87.16	0.069	83.64	84.72	0.061
	MinerU2.5 [5]	88.29	0.073	86.24	85.89	0.066
	Dolphin-v2 [3]	85.89	0.086	80.47	85.78	0.071
	PaddleOCR-VL-1.5-0.9B [1]	89.24	0.056	84.54	88.77	0.052
Dolphin-v2 [3]	MonkeyOCR [4]	81.01	0.118	81.10	73.75	0.091
	MinerU2.5 [5]	76.75	0.149	69.48	75.72	0.130
	Dolphin-v2 [3]	73.30	0.175	60.00	77.45	0.138
	PaddleOCR-VL-1.5-0.9B [1]	81.06	0.113	77.19	77.25	0.087
PP-DocLayoutV2 [2]	MonkeyOCR [4]	81.60	0.113	83.58	72.53	0.113
	MinerU2.5 [5]	81.98	0.120	84.24	73.69	0.122
	Dolphin-v2 [3]	81.11	0.130	82.56	73.75	0.126
	PaddleOCR-VL-1.5-0.9B [1]	83.28	0.102	85.49	74.55	0.107
RT-DocLayout	MonkeyOCR [4]	89.67	0.065	88.48	87.07	0.063
	MinerU2.5 [5]	90.38	0.066	90.01	87.77	0.070
	Dolphin-v2 [3]	88.61	0.082	86.30	87.71	0.079
	PaddleOCR-VL-1.5-0.9B [1]	**91.81**	**0.050**	**90.89**	**89.53**	**0.059**

Table 6: Comparison of different layout analysis modules combined with various recognition systems on Real5-OmniDocBench Skew subset. As DocLayout-YOLO does not output reading order, we employ the XYCut algorithm as a post-processing.

Layout Analysis	Recognition Method	Overall↑	Text^{Edit}↓	Formula^{CDM}↑	Table^{TEDS}↑	Reading Order^{Edit}↓
DocLayout-YOLO [7]	MonkeyOCR [4]	50.06	0.475	61.30	36.37	0.579
	MinerU2.5 [5]	46.62	0.542	56.17	37.94	0.603
	Dolphin-v2 [3]	49.71	0.492	59.62	38.72	0.587
	PaddleOCR-VL-1.5-0.9B [1]	48.74	0.493	57.82	37.66	0.583
MinerU2.5 [5]	MonkeyOCR [4]	78.56	0.184	81.69	72.41	0.123
	MinerU2.5 [5]	75.20	0.289	80.36	74.11	0.171
	Dolphin-v2 [3]	70.45	0.291	74.15	66.25	0.225
	PaddleOCR-VL-1.5-0.9B [1]	76.16	0.228	78.51	72.71	0.131
Dolphin-v2 [3]	MonkeyOCR [4]	53.12	0.333	72.56	20.09	0.184
	MinerU2.5 [5]	36.01	0.552	43.13	20.06	0.405
	Dolphin-v2 [3]	55.84	0.395	53.49	53.51	0.250
	PaddleOCR-VL-1.5-0.9B [1]	49.84	0.339	65.26	18.15	0.175
PP-DocLayoutV2 [2]	MonkeyOCR [4]	78.65	0.171	82.92	70.11	0.177
	MinerU2.5 [5]	73.44	0.276	77.43	70.46	0.230
	Dolphin-v2 [3]	77.71	0.195	82.25	70.33	0.188
	PaddleOCR-VL-1.5-0.9B [1]	77.03	0.184	79.63	69.83	0.183
RT-DocLayout	MonkeyOCR [4]	88.44	0.060	89.83	81.50	0.066
	MinerU2.5 [5]	88.06	0.083	89.26	83.18	0.071
	Dolphin-v2 [3]	88.04	0.076	87.18	84.50	0.078
	PaddleOCR-VL-1.5-0.9B [1]	**91.72**	**0.049**	**91.65**	**88.35**	**0.062**

3 Controlled Comparison with DocLayout-YOLO

To decouple the benefits of method innovation from the influence of large-scale training data, we conduct a rigorous head-to-head comparison by fine-tuning the DocLayout-YOLO [7] on our internal dataset. By synchronizing the training data and environment, we provide a transparent evaluation of the inherent advantages of RT-DocLayout.

DocLayout-YOLO [7] is trained using the official framework, starting from the provided pre-trained weights. The input resolution is fixed at 1600. For an end-to-end performance evaluation, both models employ PaddleOCR-VL-1.5-0.9B as the unified recognition engine. Notably, as DocLayout-YOLO [7] lacks native reading order prediction, we equip it with the XYCut algorithm as a post-processing step.

As demonstrated in Table 7, RT-DocLayout consistently outperforms DocLayout-YOLO [7] across all scenarios. On the OmniDocBench v1.5 [6], RT-DocLayout achieves an Overall score of 94.50, surpassing DocLayout-YOLO [7] by 2.51 points.

More crucially, RT-DocLayout exhibits remarkable robustness under the challenging conditions presented in the Real5-OmniDocBench [8] subsets. For instance, in the Warping and Skew subsets, DocLayout-YOLO [7] experiences significant performance degradation (dropping to 82.10 and 68.60, respectively), whereas RT-DocLayout maintains a stable and high Overall score (> 91.0). Specifically, our model reduces the Reading Order Edit error by over 80% in

Table 7: Fair architectural comparison between RT-DocLayout and DocLayout-YOLO [7]. Both models are fine-tuned on the same in-house dataset. As DocLayout-YOLO [7] does not output reading order, we employ the XYCut algorithm as a post-processing. "Raw" denotes the OmniDocBench v1.5 [6], while subsequent rows represent specific degradation subsets from Real5-OmniDocBench [8]. All evaluations use a fixed PaddleOCR-VL-1.5-0.9B as the recognition system.

Benchmark / Subset	Model	Overall↑	Text^{Edit}↓	Formula^{CDM}↑	Table^{TEDS}↑	Reading Order^{Edit}↓
Raw	DocLayout-YOLO [7]	91.99	0.053	90.31	90.94	0.090
	RT-DocLayout	**94.50**	**0.035**	**94.21**	**92.76**	**0.042**
Scanning	DocLayout-YOLO [7]	91.15	0.058	89.60	89.66	0.098
	RT-DocLayout	**93.25**	**0.037**	**92.33**	**91.11**	**0.046**
Warping	DocLayout-YOLO [7]	82.10	0.135	81.55	78.22	0.239
	RT-DocLayout	**91.09**	**0.053**	**90.59**	**87.95**	**0.064**
Illumination	DocLayout-YOLO [7]	89.00	0.077	87.30	87.38	0.151
	RT-DocLayout	**92.11**	**0.046**	**91.72**	**89.22**	**0.052**
Screen-Photograph	DocLayout-YOLO [7]	88.15	0.076	86.94	85.15	0.130
	RT-DocLayout	**91.81**	**0.050**	**90.89**	**89.53**	**0.059**
Skew	DocLayout-YOLO [7]	68.60	0.236	71.30	58.11	0.357
	RT-DocLayout	**91.72**	**0.049**	**91.65**	**88.35**	**0.062**

Skew compared to the DocLayout-YOLO [7] combined XYCut. These results empirically confirm that the superiority of RT-DocLayout stems from its robust design and superior spatial modeling capabilities, rather than simply relying on data scaling.

4 Visualizations and Qualitative Results

We provide additional qualitative visualization results to complement the main paper. Fig. 1 shows RT-DocLayout predictions on OmniDocBench v1.5 [6], covering a wide range of document types. Fig. 2 presents results on Real5-OmniDocBench [8] under five common real-world degradations. Across both settings, RT-DocLayout produces accurate pixel-level masks and coherent reading order, confirming its precision and robustness on diverse document layouts and capture conditions.



Fig. 1: Visualization of RT-DocLayout predictions on OmniDocBench v1.5 across diverse document types, including textbooks, academic papers, newspapers, research reports, and table-intensive documents. RT-DocLayout produces accurate pixel-level masks and reading order predictions across all document scenarios.

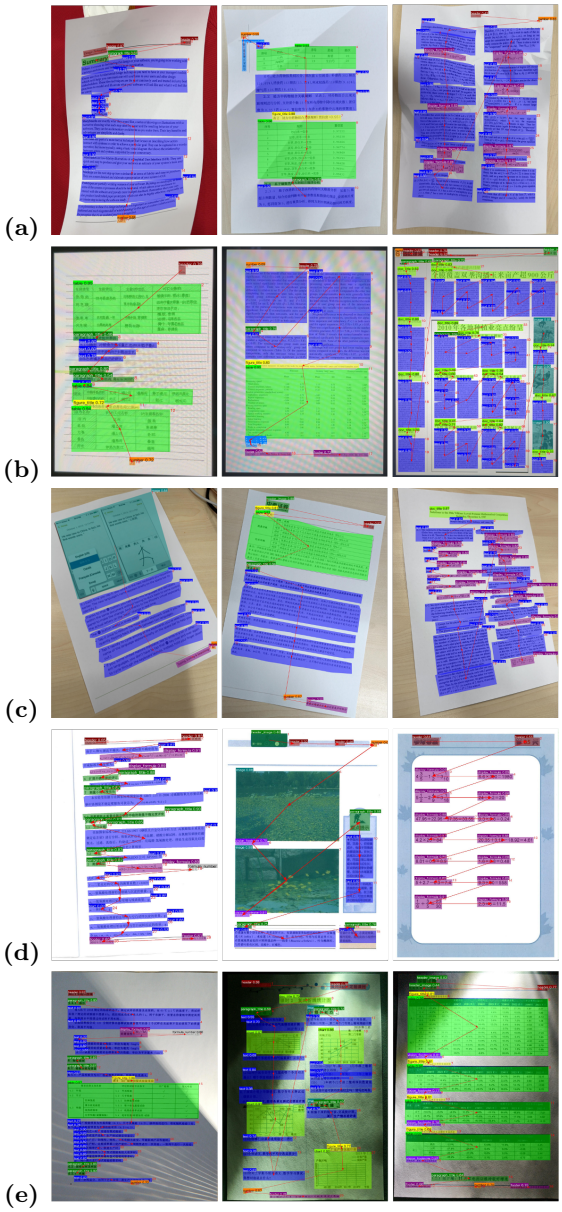


Fig. 2: Visualization of RT-DocLayout predictions on Real5-OmniDocBench under five real-world degradation scenarios. **(a)** Warping. **(b)** Screen photography. **(c)** Skew. **(d)** Scanning. **(e)** Illumination. RT-DocLayout produces accurate masks and maintains structural consistency across all conditions.

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